Graphic Era HILL UNIVERSITY

End Term(Odd)SemesterExaminationDecember2024

Roll no.....

Name of the Course and semester: B Tech Civil and 7th semester
Name of the Paper: Transportation Engineering-II (Railway & Tunnel)
Paper Code: TCE-702
Time:3 hours
Maximum Marks:100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20Marks) CO1

a. Identify the various resistances to traction on a railway track. Explain the effect of these resistances on the overall efficiency of railway operations.

b. Calculate the maximum permissible train load that can be pulled by a locomotive having pairs of driving wheels carrying an axle load of 24 tonnes each. The train has to run at a speed of 80 kmph on a straight level track (B.G). Also calculate the reduction in speed, if train has to climb a gradient of 1 in 200.

c. Evaluate the development of Indian Railways in comparison to the global railway systems in terms of infrastructure, capacity, and technological advancements

Q2. (2X10=20Marks) CO2

a. What are the main components of a railway track? Define the function of each part in ensuring the stability and safety of the track.

b. Explain the concept of creep and wear in rails.

c. What are the requirements of a good ballast? Mention the different types of ballast used in permanent way.

Q3. (2X10=20Marks) CO3

a. Describe the concept of super elevation and equilibrium cant in the design of horizontal curves.

b. A 5-degree curve diverges from a 3-degree main curve in reverse direction in the layout of a B.G yard. If the speed on branch line is restricted to 35 kmph, determine the restricted speed on the main line.

c. What would be the permissible speed on the curve? If 8' M.G. track, the average speed of different trains is 50 kmph and allowable cant deficiency is half of maximum cant deficiency.

Q4. (2X10=20Marks) CO4

a. Analyze the principle of interlocking in railway signaling systems. How does interlocking ensure safety and prevent accidents at junctions and crossings?

b. Define the concept of turnouts and explain their role in the operation of a railway track

c. A turnout is to be laid off a straight BG track with a 1 in 12 crossing. Determine the lead and radius of the turnout with the help of the following data: Heel divergence (d) = 133 mm, Crossing angle (α) = 4° 45'49", Switch angle (β) = 1°8'0", Straight length between the theoretical nose of crossing and the tangent point of crossing (h) = 1.418 m